

AS1108: Applied Physics

Introduction

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Units

<u>Fundamental Units</u>	<u>SI</u>	<u>Dimension</u>
Length	m	$[L]$
Mass	kg	$[M]$
Time	s	$[T]$
Current	A	$[I]$

<u>Derived Units</u>	<u>SI</u>	<u>Dimension</u>
Area	m^2	$[L^2]$
Speed	m/s	$[LT^{-1}]$
Acceleration	m/s^2	$[LT^{-2}]$
Force	N or $kg m/s^2$	$[MLT^{-2}]$
Energy	J or $kg m^2/s^2$	$[ML^2T^{-2}]$

Universal Constants

G - gravitational constant = $6.67 \times 10^{-11} \frac{Nm^2}{kg^2} [M^{-1}L^3T^{-2}]$

h - Planck constant = $6.626 \times 10^{-34} Js [ML^2T^{-2}]$

c - speed of light = $299792458 m/s [LT^{-1}]$

(2)

Derivative $f'(x) = \frac{df(x)}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

(i) $\frac{d}{dx} (\text{constant}) = 0$

(ii) $\frac{d}{dx} (f(x) \pm g(x)) = \frac{df(x)}{dx} \pm \frac{dg(x)}{dx}$

(iii) $\frac{d}{dx} (kf(x)) = k \frac{df(x)}{dx}$

(iv) $\frac{d}{dx} [f(x)g(x)] = f(x) \frac{dg(x)}{dx} + g(x) \frac{df(x)}{dx}$
 $\rightarrow = f(x)g'(x) + g(x)f'(x)$

(v) $\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}$

(vi) $\frac{d}{dx} [f(g(x))] = f'(g(x)) \cdot g'(x)$

(vii) $\frac{d}{dx} x^n = n x^{n-1}$

(viii) $\frac{d}{dx} \cos x = -\sin x$

(ix) $\frac{d}{dx} \sin x = \cos x$

(x) $\frac{d}{dx} \tan x = \sec^2 x$

(xi) $\frac{d}{dx} \cot x = -\operatorname{cosec}^2 x$

(xii) $\frac{d}{dx} \sec x = \sec x \cdot \tan x$

(xiii) $\frac{d}{dx} \operatorname{cosec} x = -\operatorname{cosec} x \tan x$

(xiv) $\frac{d}{dx} e^x = e^x$

(xv) $\frac{d}{dx} (\ln x) = \frac{1}{x}$

(xvi) $\frac{d}{dx} a^x = a^x \ln a$

(xvii) $\frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}}$

(xviii) $\frac{d}{dx} \cos^{-1} x = \frac{-1}{\sqrt{1-x^2}}$

(xix) $\frac{d}{dx} \tan^{-1} x = \frac{1}{x^2+1}$

(xx) $\frac{d}{dx} \cot^{-1} x = \frac{-1}{x^2+1}$

(xxi) $\frac{d}{dx} \sec^{-1} x = \frac{1}{|x| \sqrt{x^2-1}}$

(xxii) $\frac{d}{dx} \operatorname{cosec}^{-1} x = \frac{-1}{|x| \sqrt{x^2-1}}$

Chain rule $\frac{df(x)}{dt} = \frac{df(x)}{dx} \cdot \frac{dx}{dt}$

Integrals

(i) $\int dx = x + C$

(ii) $\int x^n dx = \frac{x^{n+1}}{n+1} + C$

(iii) $\int \frac{dx}{x} = \ln|x| + C$

(iv) $\int e^x dx = e^x + C$

(v) $\int a^x dx = \frac{1}{\ln a} a^x + C$

(vi) $\int \ln x dx = x \ln x - x + C$

(vii) $\int \sin x dx = -\cos x + C$

(viii) $\int \cos x dx = \sin x + C$

(ix) $\int \tan x dx = -\ln|\cos x| + C$

(x) $\int \cot x dx = \ln|\sin x| + C$

(xi) $\int \sec x dx = \ln|\sec x + \tan x| + C$

(xii) $\int \csc x dx = -\ln|\csc x + \cot x| + C$

(xiii) $\int \sec^2 x dx = \tan x + C$

(xiv) $\int \operatorname{cosec}^2 x = -\cot x + C$

(xv) $\int \sec x \tan x dx = \sec x + C$

(xvi) $\int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + C$

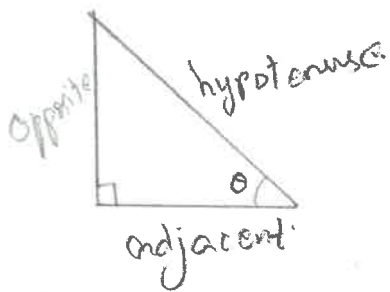
(xvii) $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C$

(xviii) $\int \frac{dx}{a^2 + x^2} = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + C$

(xix) $\int \frac{dx}{x \sqrt{x^2 - a^2}} = \frac{1}{a} \sec^{-1}\left(\frac{|x|}{a}\right) + C$

Integration by parts

$$\int u dv = uv - \int v du$$



$$\sin \theta = \frac{\text{opp}}{\text{hyp}}$$

$$\cos \theta = \frac{\text{adj}}{\text{hyp}}$$

$$\begin{aligned} \tan \theta &= \frac{\text{opp}}{\text{adj}} \\ &= \frac{\sin \theta}{\cos \theta} \end{aligned}$$

$$\csc \theta = \frac{\text{hyp}}{\text{opp}} = \frac{1}{\sin \theta} \quad (2)$$

$$\sec \theta = \frac{\text{hyp}}{\text{adj}} = \frac{1}{\cos \theta}$$

$$\begin{aligned} \cot \theta &= \frac{\text{adj}}{\text{opp}} = \frac{1}{\tan \theta} \\ &= \frac{\cos \theta}{\sin \theta} \end{aligned}$$

$$(i) \sin(-\theta) = -\sin \theta$$

$$(ii) \cos(-\theta) = \cos \theta$$

$$(iii) \sin\left(\frac{\pi}{2} - \theta\right) = \cos \theta$$

$$(iv) \cos\left(\frac{\pi}{2} - \theta\right) = \sin \theta$$

$$(v) \sin(\pi - \theta) = \sin \theta$$

$$(vi) \sin(\pi + \theta) = -\sin \theta$$

$$(vii) \sin 2\theta = 2 \sin \theta \cos \theta$$

$$\begin{aligned} (viii) \cos 2\theta &= \cos^2 \theta - \sin^2 \theta \\ &= 2 \cos^2 \theta - 1 \\ &= 1 - 2 \sin^2 \theta \end{aligned}$$

$$(ix) \sin^2 \theta + \cos^2 \theta = 1$$

$$(x) \tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

θ	0°	30°	45°	60°	90°
$\sin$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞

$$\sin 180^\circ = 0$$

$$\tan 180^\circ = 0$$

$$\cos 180^\circ = -1$$